

(2) **Solution:** $(A-B) \cap (B-C) \cap (A-C) = (A \cap \bar{B}) \cap (B \cap \bar{C}) \cap (A \cap \bar{C})$ (Prob-21(a))

$$\begin{aligned}
 &= ((A \cap \bar{B}) \cap (B \cap \bar{C})) \cap (A \cap \bar{C}) \\
 &= ((A \cap \bar{B}) \cap B) \cap \bar{C} \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= (A \cap (\bar{B} \cap B)) \cap \bar{C} \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= (A \cap (\bar{B} \cap B)) \cap \bar{C} \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= ((A \cap (\bar{B} \cap B)) \cap \bar{C}) \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= ((A \cap \Phi) \cap \bar{C}) \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Complement Law}) \\
 &= (\Phi \cap \bar{C}) \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Domination Law}) \\
 &= (\bar{C} \cap \Phi) \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= \Phi \cap (A \cap \bar{C}) \quad (2^{\text{nd}} \text{ Complement Law}) \\
 &= (A \cap \bar{C}) \cap \Phi \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= \Phi \quad (2^{\text{nd}} \text{ Complement Law}) \quad \square
 \end{aligned}$$

35) Let A, B and C be sets. Use the identities in Table 1 to show that

$$(\overline{A \cup B}) \cap (\overline{B \cup C}) \cap (\overline{A \cup C}) = \overline{A \cap B \cap C}$$

Solution:

$$\begin{aligned}
 &(\overline{A \cup B}) \cap (\overline{B \cup C}) \cap (\overline{A \cup C}) = (\overline{A \cap B}) \cap (\overline{B \cap C}) \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ De Morgan's Law}) \\
 &= ((\overline{A \cap B}) \cap (\overline{B \cap C})) \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= ((\overline{A \cap B}) \cap \bar{B}) \cap \bar{C} \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= (\overline{A \cap B} \cap \bar{B}) \cap \bar{C} \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ Idempotent Law}) \\
 &= (\overline{A \cap B} \cap \bar{C}) \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= (\overline{A \cap B}) \cap \bar{C} \cap (\overline{A \cap C}) \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= ((\overline{A \cap B}) \cap \bar{C}) \cap \bar{A} \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= ((\overline{A \cap B}) \cap \bar{C} \cap \bar{A}) \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= (\overline{A \cap B} \cap \bar{C}) \cap \bar{A} \quad (2^{\text{nd}} \text{ Idempotent Law}) \\
 &= (\overline{A \cap B} \cap \bar{C}) \cap \bar{A} \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= (\overline{A \cap B} \cap \bar{C}) \cap \bar{A} \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= \overline{A \cap B \cap C} \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= \overline{A \cap B \cap C} \quad (2^{\text{nd}} \text{ Idempotent Law}) \\
 &= \overline{A \cap B \cap C} \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= \overline{A \cap B \cap C} \quad \square
 \end{aligned}$$